CENG-543 Research Paper Presentation

ALİ EKİN ÖZÇETİN

GENERALIZATION THROUGH MEMORIZATION: NEAREST NEIGHBOR LANGUAGE MODELS

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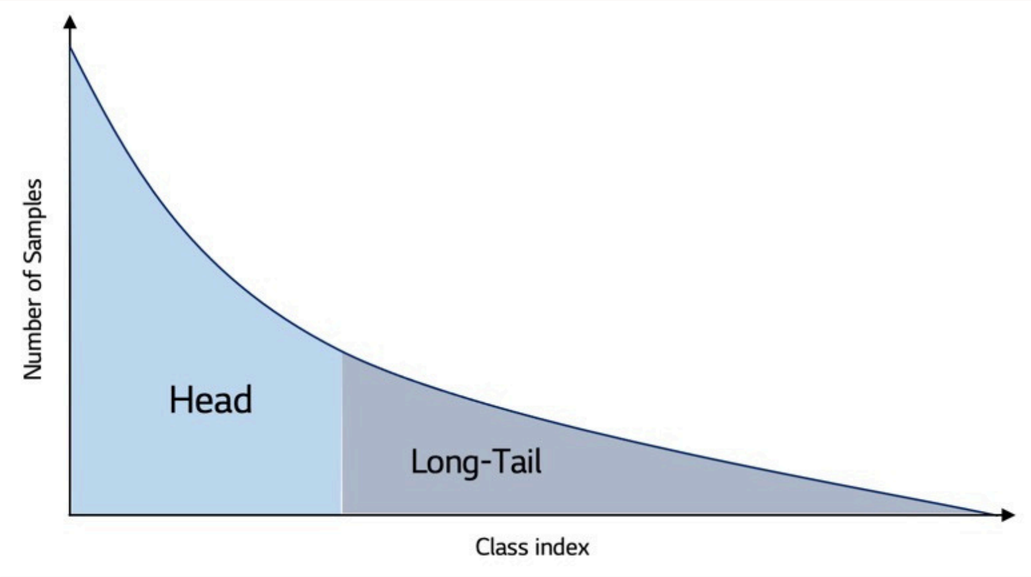
PUBLİSHED AT: ICLR 2020

The Problem – The Dilemma of Language Models

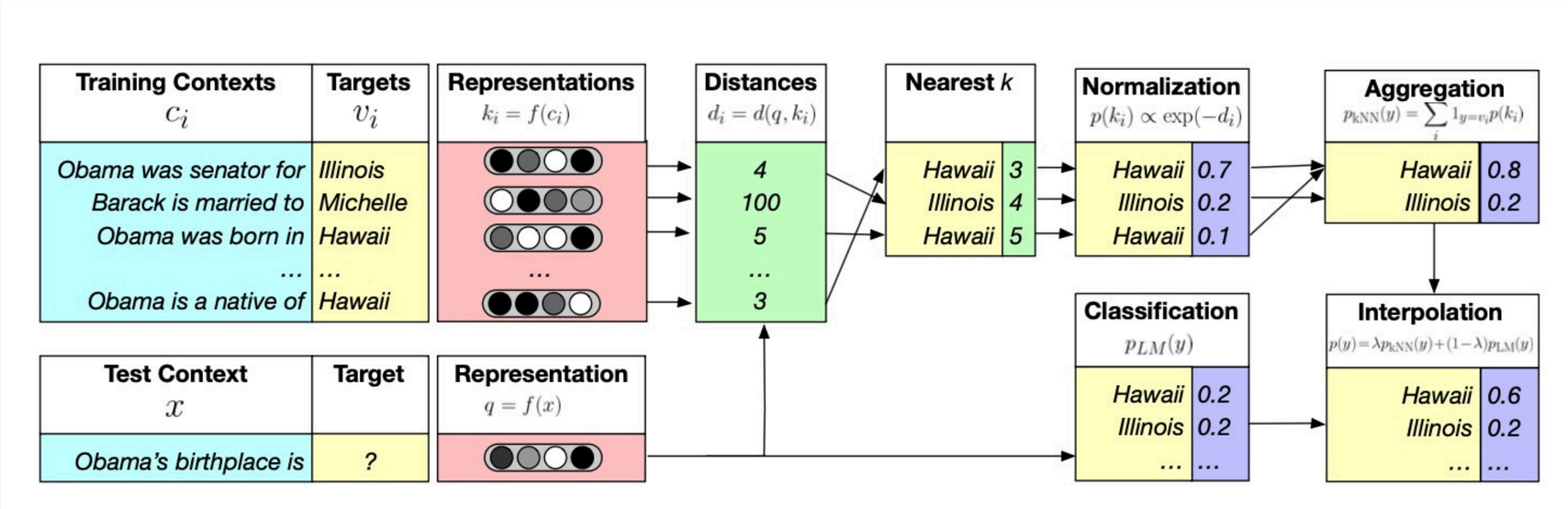
- **Simultaneous Tasks:** LMs must simultaneously map prefixes to fixed-size representations and predict the next word using those vectors.
- **The Syntax-Fact Gap:** While models learn syntax and language structure effectively, they struggle to encode rare, factual details (names, dates) within fixed parameters.
- **Long-Tail Failure:** Generalization is strong for frequent patterns but fails in the "long tail" of world knowledge.
- **The Storage Problem:** Traditional models try to store all facts inside their internal weights. This is inefficient because to learn more facts, the model must become much larger.



The Core Hypothesis

- **The Main Idea:** It is easier to see that two sentences are similar than to guess the exact next word.
- **The Trade-off:** Standard models try to do two things at once: understand the language and remember every fact. This is very difficult for a single network.
- **New Approach:** Instead of making the model bigger, we separate thinking from memory:
  - **Base Model (Thinking):** Focuses on understanding the meaning of the sentence.
  - **kNN (Memory):** An external database that stores facts and rare details.
- **The Result:** The model can "know" more things by just increasing the database size, without needing more parameters.

Methodology - The kNN-LM Architecture



- **Core Idea:** The kNN-LM augments a pretrained language model with a k-nearest neighbor retrieval mechanism that acts as an external memory.
- **How It Works:**
- **1- Datastore Construction:**
  - Built offline from the training corpus.
  - Each entry stores a (key, value) pair:
  - **key (k_i):** Context representation $f(c_i)$ from the final Transformer layer.
  - **value (v_i):** The target word following that context.

- **2- Inference (Test Time):**
  - The model encodes the current context into a vector f(x).
  - Using FAISS (Johnson et al., 2017), it retrieves the k most similar contexts by L_2 distance.
  - A neighbor distribution $p_{knn}(y|x)$ is computed via softmax over distances.
- **3- Interpolation:**
  - $p(y \mid x) = \lambda p_{kNN}(y \mid x) + (1-\lambda) p_{LM}(y \mid x)$
  - Here, λ controls how much the model relies on retrieved memory versus its own learned distribution.
- This mechanism gives the LM a lookup-based memory , it doesn't just guess the next word, it retrieves it from experience.

Experimental Setup

- **Datasets:**
  - **WikiText-103:** High-quality Wikipedia articles (~100M tokens) used as a standard benchmark.
  - **Toronto Books Corpus:** Diverse fictional texts used for domain adaptation testing.
  - **WIKI-3B & WIKI-100M:** Larger Wikipedia datasets (up to 3B tokens) used to test how the model scales with more memory
- **Base Model:**
  - Transformer LM (Baevski & Auli, 2019) with adaptive input representations.
  - Pretrained on WikiText-103; later combined with external datastore.
- **Implementation:**
  - Framework: PyTorch.
  - Retrieval system: FAISS for efficient nearest-neighbor search over billions of vectors.
  - Datastore built from hidden representations (context embeddings) of the final Transformer layer.
- **Evaluation Metric:**
  - Perplexity (PPL) - lower is better, measuring model's predictive confidence.
  - Compared across datasets to assess both in-domain and out-of-domain performance.

Using Training Data As The Datastore

1. Primary Results: WikiText-103 (In-Domain)

- kNN-LM achieves state-of-the-art (SOTA) results on the WikiText-103 benchmark without any additional training or fine-tuning of the base model.

Model	Perplexity (↓) Dev	Perplexity (↓) Test	# Trainable Params
Baevski & Auli (2019)	17.96	18.65	247M
+Transformer-XL (Dai et al., 2019)	-	18.30	257M
+Phrase Induction (Luo et al., 2019)	-	17.40	257M
Base LM (Baevski & Auli, 2019)	17.96	18.65	247M
+kNN-LM	**16.06**	**16.12**	247M
+Continuous Cache (Grave et al., 2017c)	17.67	18.27	247M
+kNN-LM + Continuous Cache	**15.81**	**15.79**	247M

2. Generalization: Books Corpus (Cross-Domain)

- The method remains highly effective even when applied to different domains, proving that the retrieval mechanism generalizes well beyond the original training distribution.

Model	Perplexity (↓) Dev	Perplexity (↓) Test	# Trainable Params
Base LM (Baevski & Auli, 2019)	14.75	11.89	247M
+kNN-LM	**14.20**	**10.89**	247M

3. Key Findings

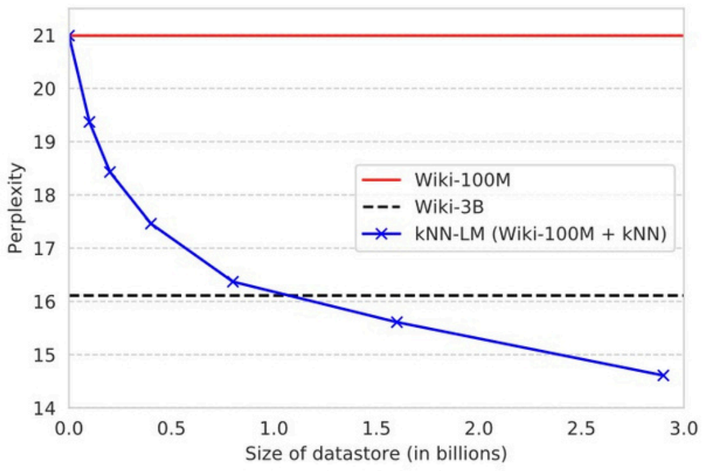
- **Memory vs. Learning:** Performance improves simply by accessing external memory (the datastore) rather than updating the model's internal weights.
- **Cross-Domain Success:** The Books Corpus results show that the model can adapt to new distributions simply by swapping or adding to the datastore.
- **Efficiency:** This method significantly improves the base model without adding any new trainable parameters, saving time and computational resources.

More Data Without Training

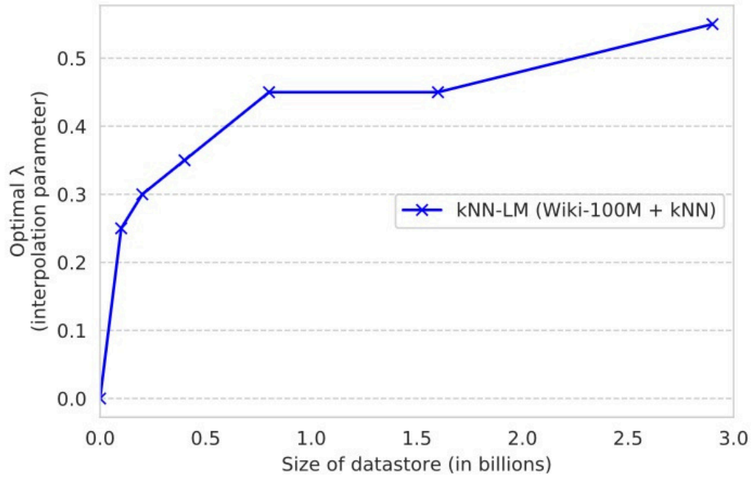
Training Data	Datastore	Perplexity (↓) Dev	Test
WIKI-3B	-	16.11	15.17
WIKI-100M	-	20.99	19.59
WIKI-100M	WIKI-3B	14.61	13.73

- **Efficiency:** A small LM (100M) + large datastore (3B) outperforms a fully trained 3B-token LM.
- **Expandable Memory:** Increase capacity by adding data, without retraining or parameter growth.
- **New Paradigm:** Performance scales by increasing memory size rather than making the model larger.

Effect of Datastore Size and Interpolation Parameter (λ)



(a) Effect of datastore size on perplexities.



(b) Tuned values of λ for different datastore sizes.

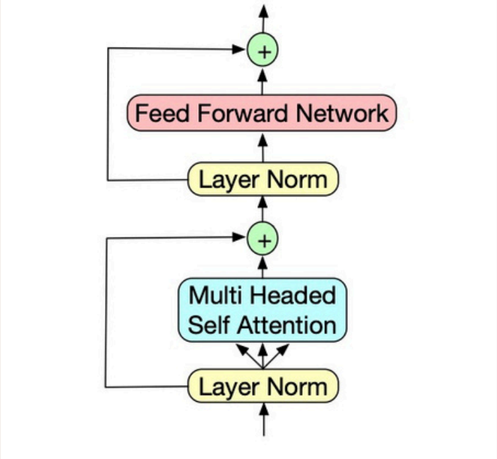
- **Continuous Improvement:** Performance scales with datastore size; no saturation seen even at 3B tokens.
- **Memory > Training:** A small LM (100M) + 1.6B datastore outperforms a model fully trained on 3B tokens.
- **The λ Factor:** As the datastore grows, the optimal λ increases—the model relies more on retrieval than its internal weights.
- **Substitution:** Access to more memory can effectively substitute for more intensive training.

Domain Adaptation on the Books Dataset

Training Data	Datastore	Perplexity (↓) Dev	Test
WIKI-3B	-	37.13	34.84
BOOKS	-	14.75	11.89
WIKI-3B	BOOKS	24.85	20.47

- **Zero Fine-Tuning:** kNN-LM adapts to new domains without any retraining or parameter updates.
- **Domain Mismatch:** A Wikipedia-trained model performs poorly on the Books dataset (PPL: 34.84).
- Simply swapping the Wikipedia datastore for a Books datastore drops Perplexity to 20.47.
- **Competitiveness:** This result nearly matches the performance of models specifically trained on the Books domain.
- **Efficiency:** Shows that adapting memory is a lightweight, instant alternative to expensive model retraining.

Transformer Representations and Memory Choice

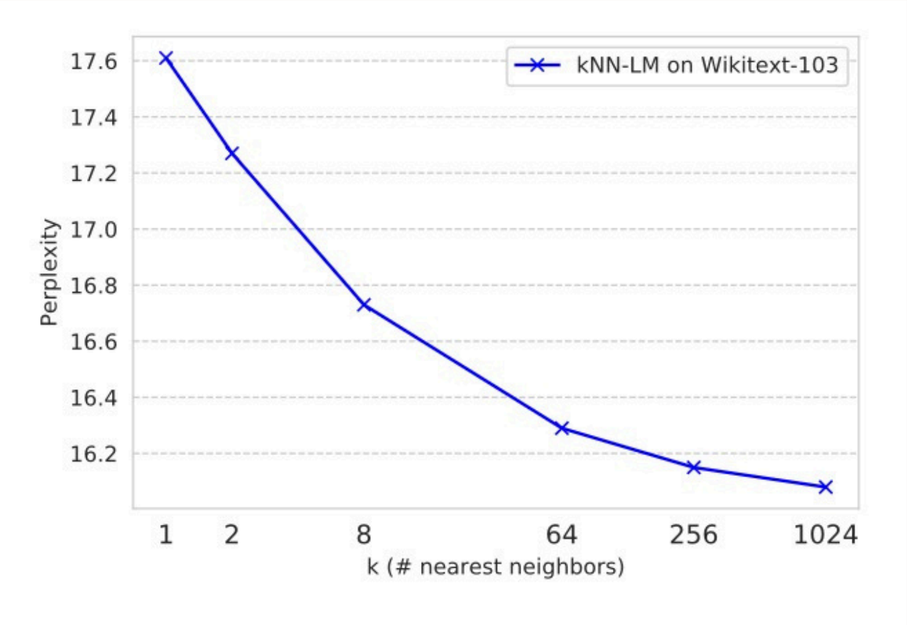


Key Type	Dev ppl. (↓)
No datastore	17.96
Model output	17.07
Model output layer normalized	17.01
FFN input after layer norm	**16.06**
FFN input before layer norm	17.06
MHSA input after layer norm	16.76
MHSA input before layer norm	17.14

- **The Goal:** Identify which Transformer layer provides the most effective representation for memory retrieval.
- **Component Testing:** Evaluated points across LayerNorm, Self-Attention, and Feed-Forward (FFN) blocks.
- Extracting the FFN input after LayerNorm yielded the best results (PPL: 16.06).
- Representations from earlier or later layers performed significantly worse (≈17–18).
- **Key Insight:** This specific stage captures the perfect balance of syntax and semantics before the final prediction.

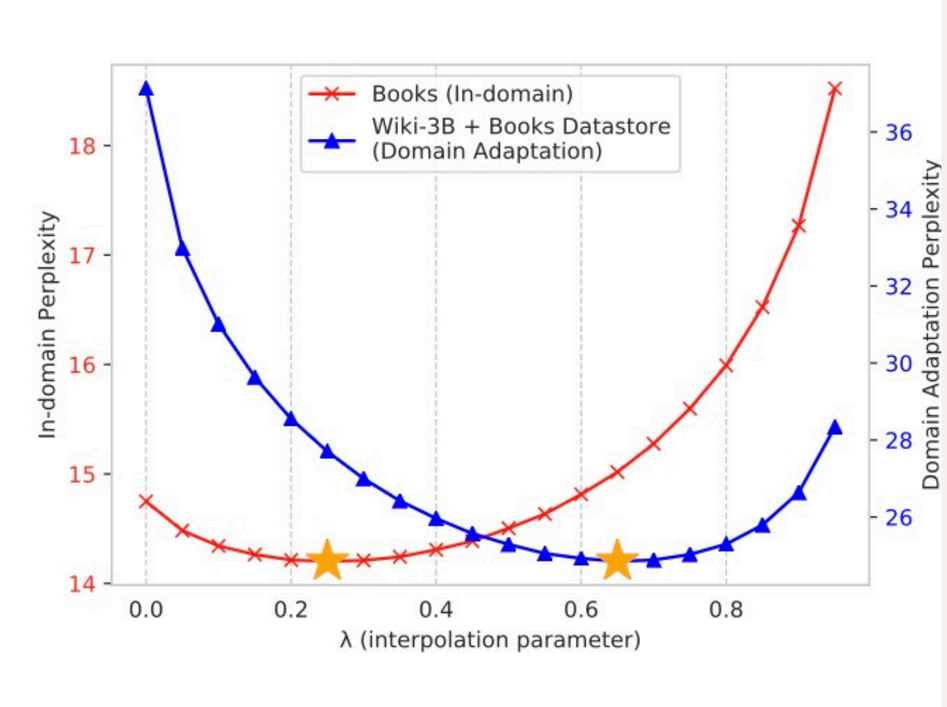
Effect of the Number of Nearest Neighbors (k)

- **Monotonic Improvement:** Performance on WikiText-103 improves consistently as the number of retrieved neighbors (k) increases.
- **Contextual Evidence:** A larger k allows the model to consider more diverse contextual evidence, resulting in lower Perplexity.
- **The Trade-off:** While accuracy improves, higher k values increase inference latency due to the computational cost of processing more lookups.
- **Efficiency:** Strong results are achieved even with moderate values (e.g., k=1024), demonstrating a high return on memory efficiency.
- **Key Finding:** Retrieval depth (k) is just as critical as datastore size for maximizing prediction accuracy.



Effect of the Interpolation Parameter (λ) on Domain Adaptation

- **Balancing Act:** The λ parameter determines the weight of retrieved memory versus the model's internal predictions.
- **Out-of-Domain Advantage:** Increasing λ improves performance on unfamiliar data, as the model relies more on the external datastore for context.
- **In-Domain Precision:** Smaller λ values are better for familiar data where the LM's internal learned knowledge is already highly reliable.
- **Dynamic Balance:** λ serves as a critical bridge between generalization (internal weights) and memorization(external retrieval).



Qualitative Analysis: Factual Retrieval

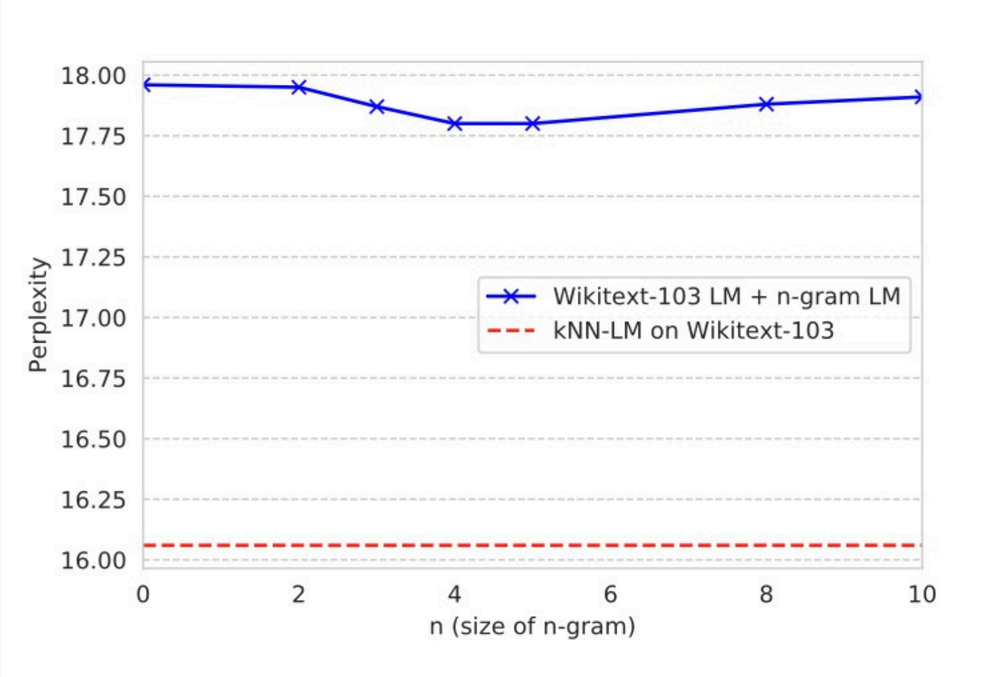
- **Base LM Struggle:** The standard model assigns very low confidence to the correct next word: "development".
- **Memory Retrieval:** kNN-LM searches the datastore and finds a nearly identical training context.
- **Result:** By leveraging this retrieved neighbor, the model assigns a massive 0.998 probability to the correct target.
- **Conclusion:** This demonstrates kNN-LM's ability to recall specific factual information directly from memory, acting as a high-precision lookup table.

Test Context ($p_{\text{kNN}} = 0.998$, $p_{\text{LM}} = 0.124$)	**Test Target**
it was organised by New Zealand international player Joseph Warbrick, promoted by civil servant Thomas Eyton, and managed by James Scott, a publican. The Natives were the first New Zealand team to perform a haka, and also the first to wear all black. They played 107 rugby matches during the tour, as well as a small number of Victorian Rules football and association football matches in Australia. Having made a significant impact on the...	development

Training Set Context	**Training Set Target**	**Context Probability**
As the captain and instigator of the 1888-89 Natives – the first New Zealand team to tour the British Isles – Warbrick had a lasting impact on the...	development	0.998
promoted to a new first grade competition which started in 1900. Glebe immediately made a big impact on the...	district	0.00012
centuries, few were as large as other players managed. However, others contend that his impact on the...	game	0.000034
Nearly every game in the main series has either an anime or manga adaptation, or both. The series has had a significant impact on the...	development	0.00000092

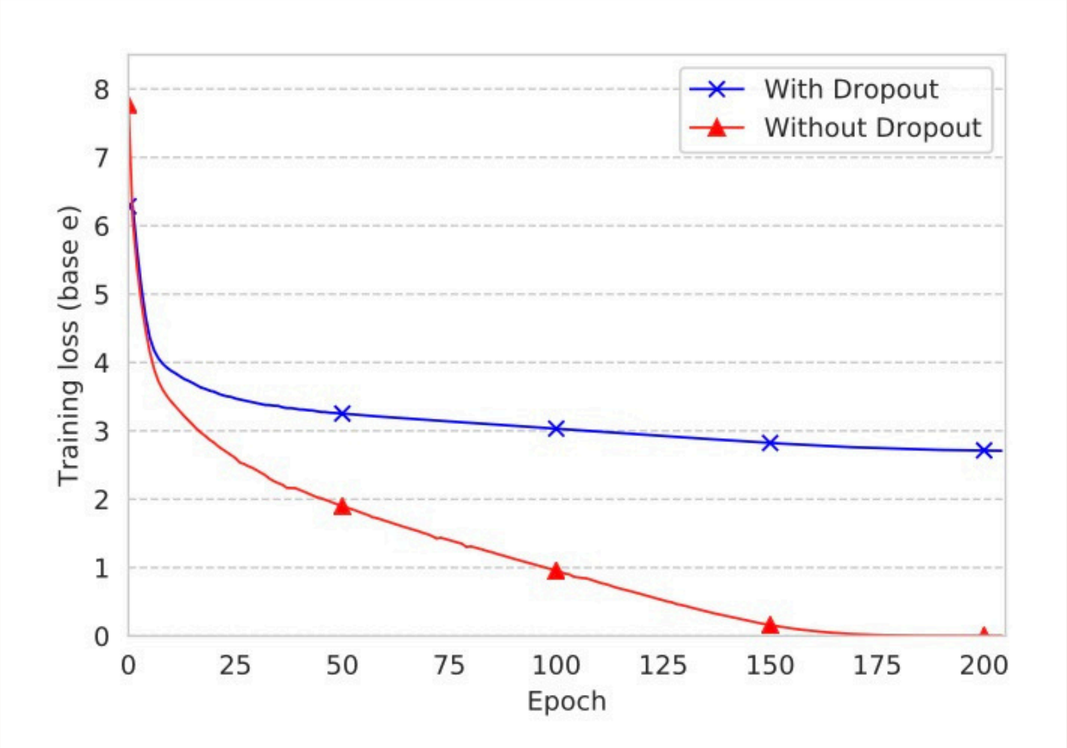
kNN-LM vs. n-gram Interpolation

- **Performance Gap:** Despite increasing n, the n-gram baseline plateaus at 17.7 Perplexity, failing to catch up.
- **The kNN Advantage:** kNN-LM achieves significantly lower error rates (~16.0 Perplexity), maintaining a clear lead.
- Unlike n-grams which rely on exact word sequences, kNN-LM uses dense contextual representations.
- **Conclusion:** The model captures deep semantic and syntactic patterns that simple word co-occurrence cannot match.



Training Behavior: Implicit vs. Explicit Memory

- **Implicit Memorization Fails:** A Transformer trained without dropout can memorize the entire dataset (Training Loss ≈0) but fails to generalize (Validation Perplexity ≈ 28.59).
- **Ineffective Interpolation:** Combining this "memorizing model" with the base LM yields negligible improvement (+0.1), unlike the significant gain from kNN (+1.9).
- **Conclusion:** Explicit memory (kNN) allows the system to memorize training data without degrading the generalization capability of the neural network.



Limitations & Future Work

Challenges

- **Latency:** Retrieval adds significant computational overhead. Searching billions of vectors for every single token is slow.
- **Storage:** Large-scale datastores require massive memory footprints, often reaching terabytes of space.
- **Privacy Risk:** Storing raw training data in the datastore poses a risk of exposing sensitive or personally identifiable information (PII).

Future Work

- **Efficiency:** Developing faster, approximate retrieval methods to reduce inference time.
- **Compression:** Exploring compressed or "learned" datastores to reduce the storage burden.
- **Hybrid Memory:** Investigating systems that combine neural recall with symbolic reasoning for better performance.

Related Work

- **Evolution of Memory:**
  - **Cache-based LMs:** Reuse recent context but rely on parameter updates.
  - **Differentiable Memory:** Models learn memory during training; effective but computationally expensive.
- **Inspiration from CV:** kNN methods in Computer Vision improved robustness, inspiring similar retrieval mechanisms for NLP.
- **Early NLP Systems:** Previous kNN systems used for tagging/morphology lacked deep learned representations.
- **The kNN-LM Edge:** Retrieves individual tokens rather than full sentences.
  - Keeps memory static and non-parametric (no retraining needed).
- **New Paradigm:** Shifts focus from "Massive Model + Parameters" to "Small Model + Large Datastore."

Conclusion

- Memorization is not the enemy of generalization; explicit memory (kNN) enhances implicit neural understanding.
- By decoupling memory (datastore) from computation (parameters), models generalize better without expensive retraining.
- **Hybrid Learning:** Successfully bridges parametric learning (neural weights) and non-parametric retrieval (external memory).
- **The Foundation of RAG:** This work established the groundwork for modern Retrieval-Augmented Generation (RAG)architectures.
- **A Key Limitation:** The model relies on static corpora; adapting to rapidly evolving, real-time information remains a challenge.
- The future of AI is not just about building bigger models, but about building models that remember smarter.

THANKS FOR LISTENING